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The thermal history of the IGM

Jamie Bolton

with thanks to

George Becker, Elisa Boera, Antonella Garzilli, Bradley Greig,
Martin Haehnelt, Michael Murphy, Matteo Viel, Stuart Wyithe

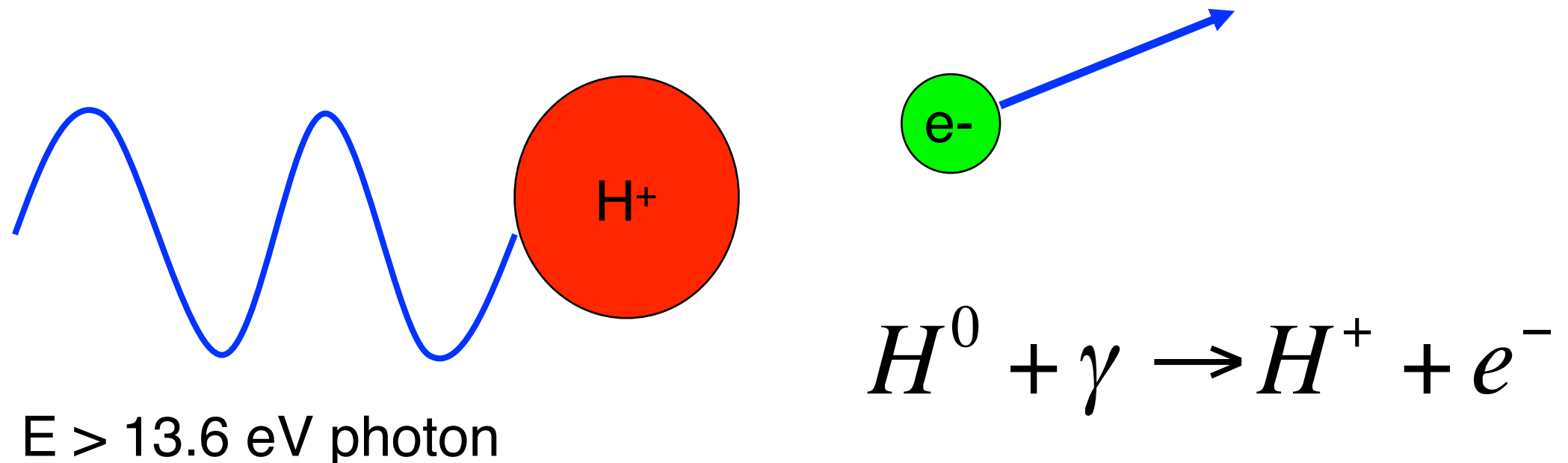


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Photo-ionisation heating



Ejected photo-electrons share their energy with neutrals via scattering and raise the temperature of the residual H-I.

Photo-ionisation heating

Low density ($\Delta < 10$), highly ionised IGM in photo-ionisation equilibrium

$$\frac{dT}{dt} = \frac{2}{3k_B} \langle E \rangle \alpha(T) n - 2HT$$

Miralda-Escudé & Rees (1994)

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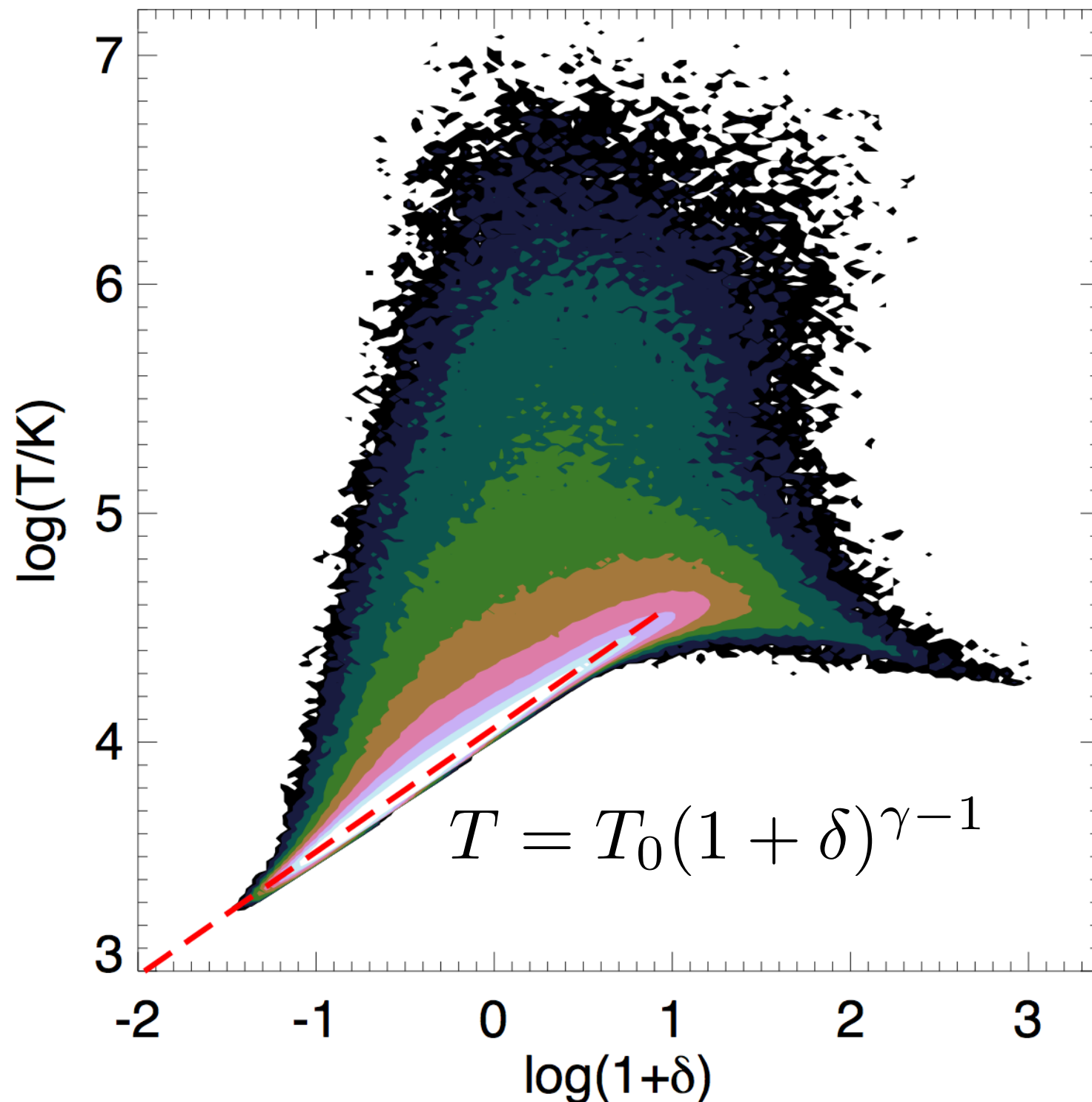
Miralda-Escudé & Rees (1994)

Optically thin IGM, power-law spectrum for UV background: $J_\nu \propto \nu^{-\beta}$

$$\langle E \rangle = \frac{h\nu_i}{\beta + 2}$$

Abel & Haehnelt (1999)

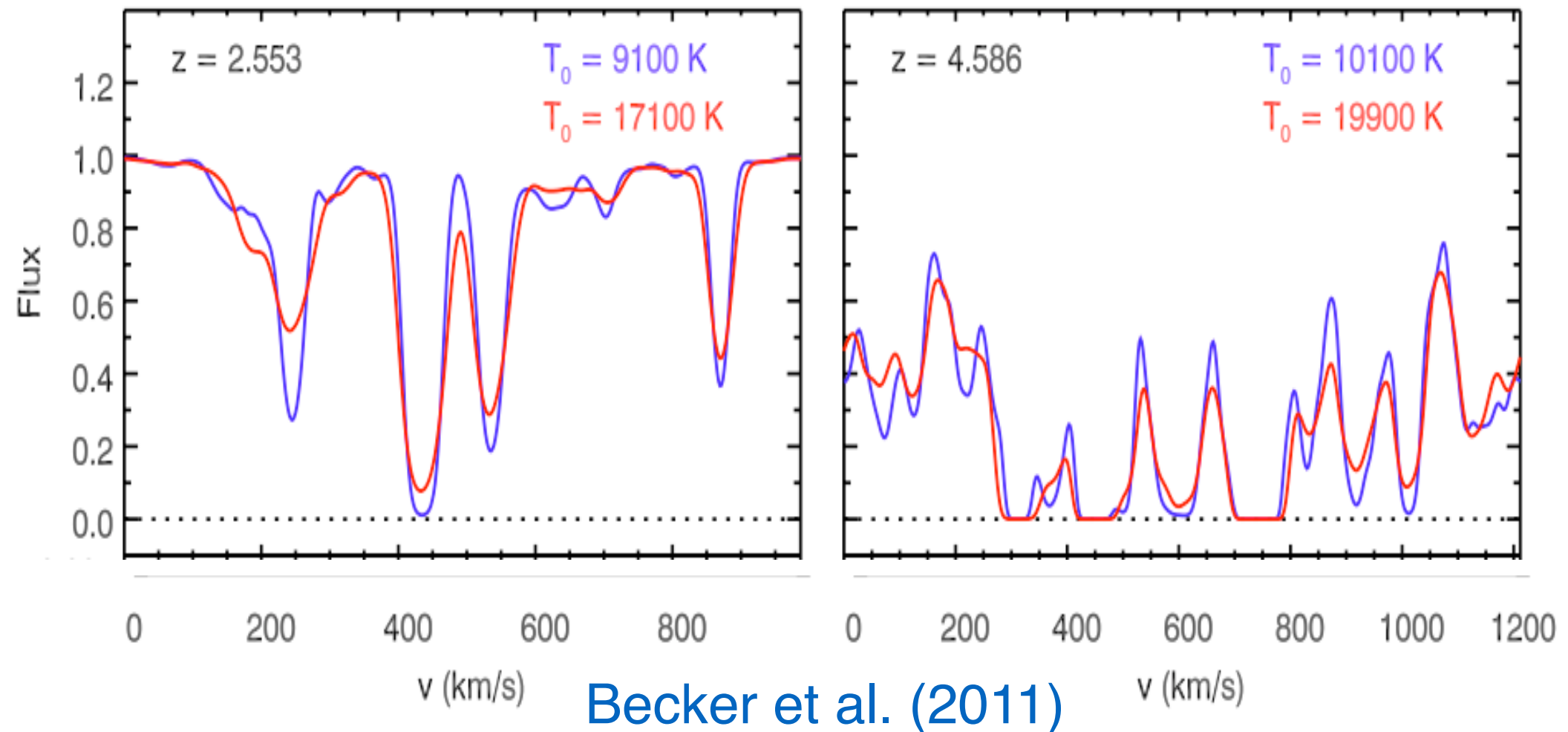
The temperature-density relation



Optically thin IGM,
power-law relationship
between temperature
and density, $\gamma \sim 1.0-1.6$

Hui & Gnedin (1997),
Valageas et al. (2002)

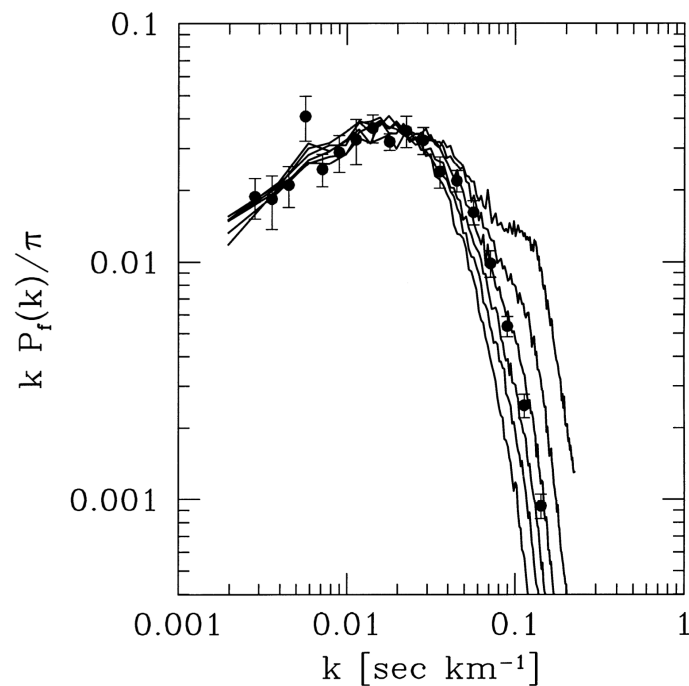
The Ly- α forest as a thermometer



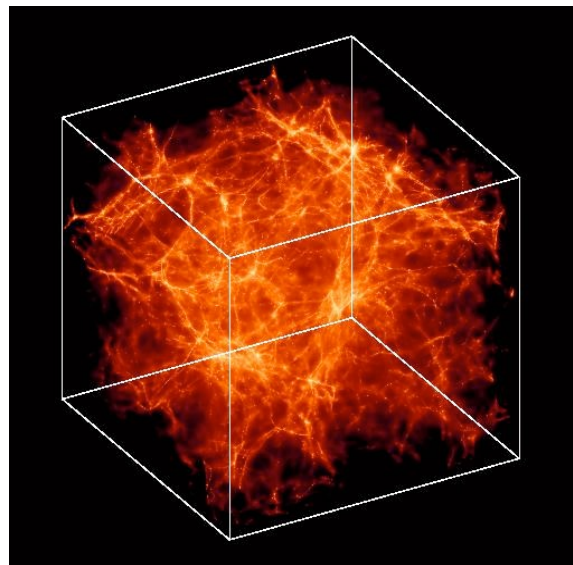
- 1) Thermal broadening by instantaneous temperature (along the line of sight only);
- 2) Jeans smoothing via integrated heating history (in three dimensions).

Measuring the temperature

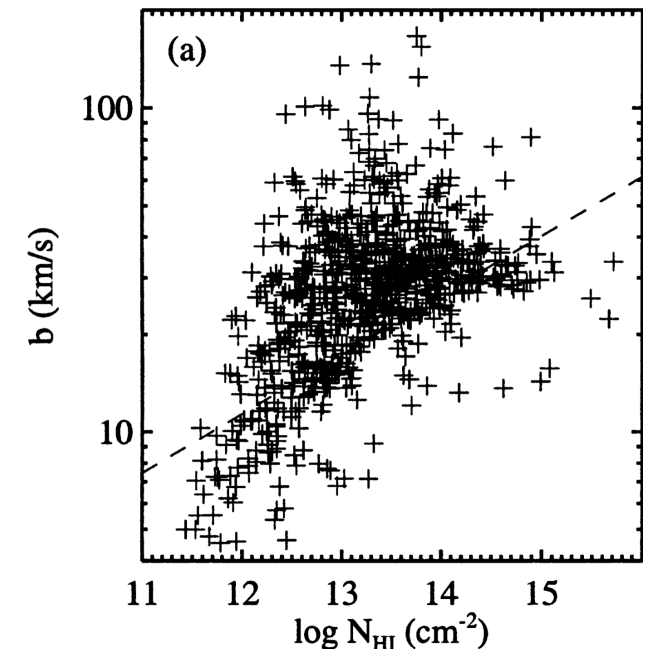
Flux statistics



IGM model



Line decomposition

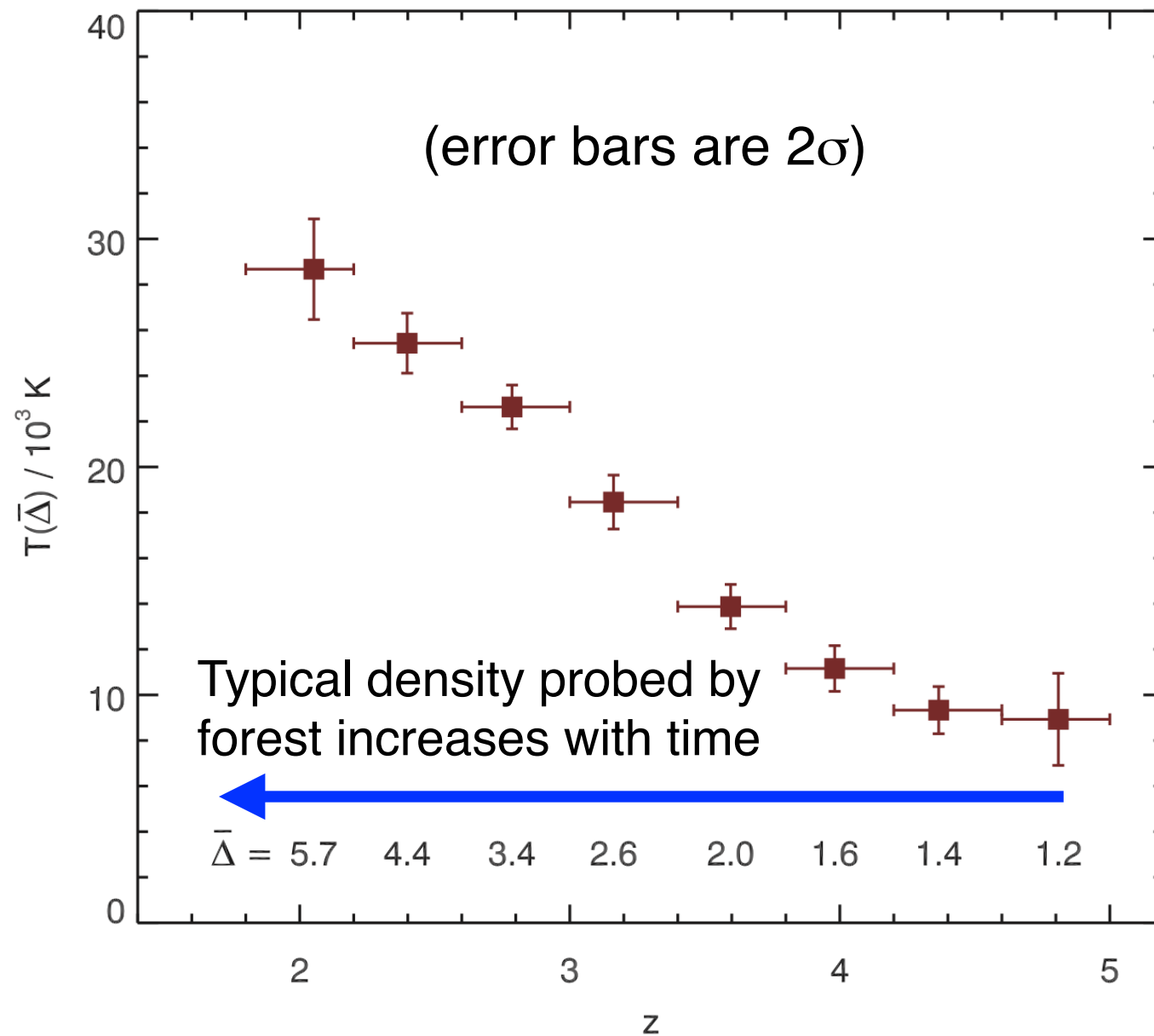


Zaldarriaga et al. (2001) [P(k)]
 Theuns et al. (2002) [wavelets]
 Bolton et al. (2008) [PDF]
 Viel et al. (2009) [P(k)+PDF]
 Lidz et al. (2010) [wavelets]
 Becker et al. (2011) [curvature]
 Garzilli et al. (2012) [wavelets]
 Calura et al. (2012) [PDF]
 Boera et al. (2014) [curvature]
 Lee et al. (2014) [PDF]

T_0, γ

Schaye et al. (2000)
 Ricotti et al. (2000)
 Bryan & Machacek (2000)
 McDonald et al. (2001)
 Bolton et al. (2012)
 Rudie et al. (2012)
 Bolton et al. (2014)

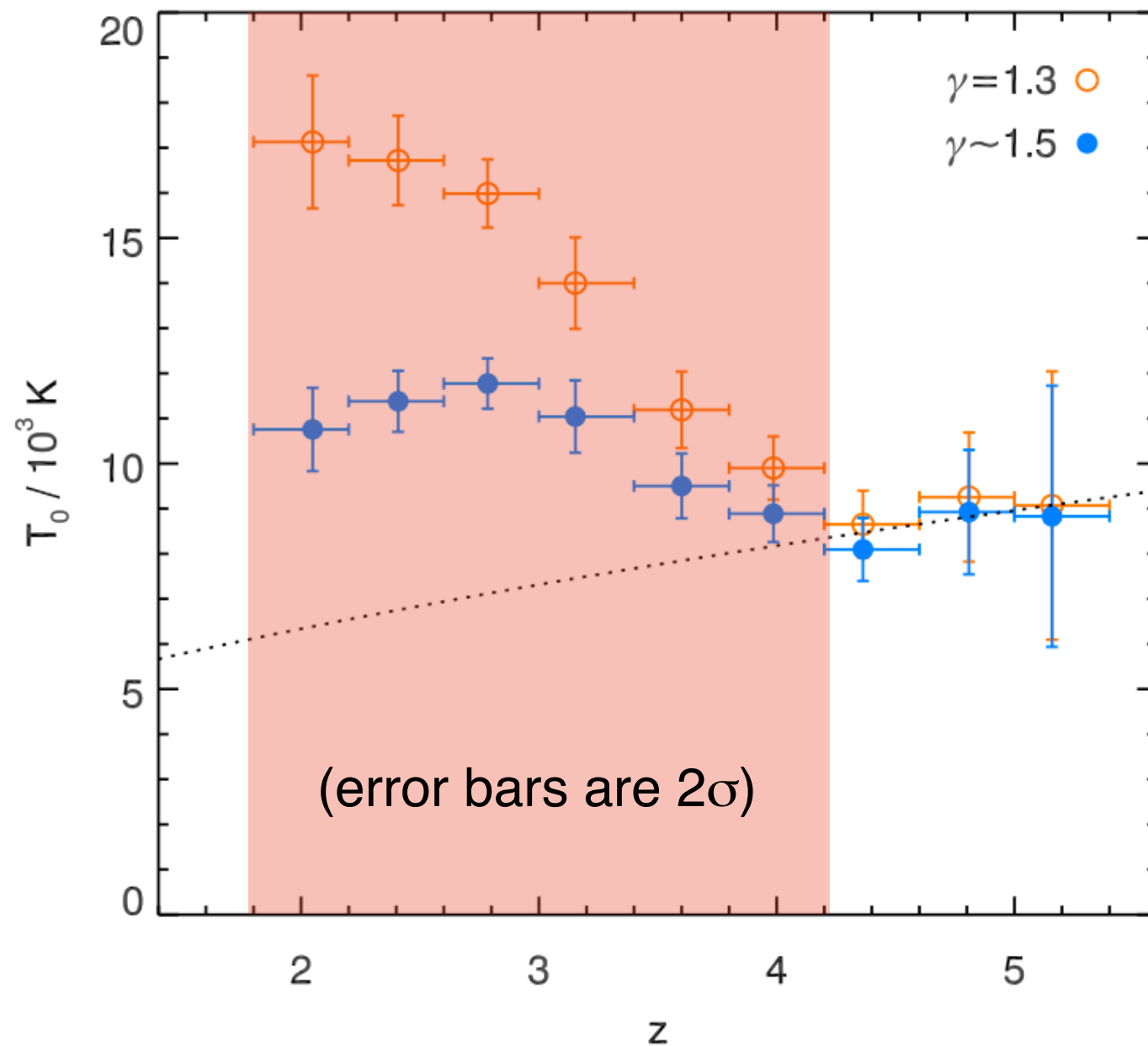
The temperature evolution



$T(\Delta)$ probed by forest well measured from **curvature** at $2 < z < 5$ in 61 HIRES and MIKE spectra ([Elisa's talk next](#))

Becker et al. (2011),
see also Boera et al. (2014)

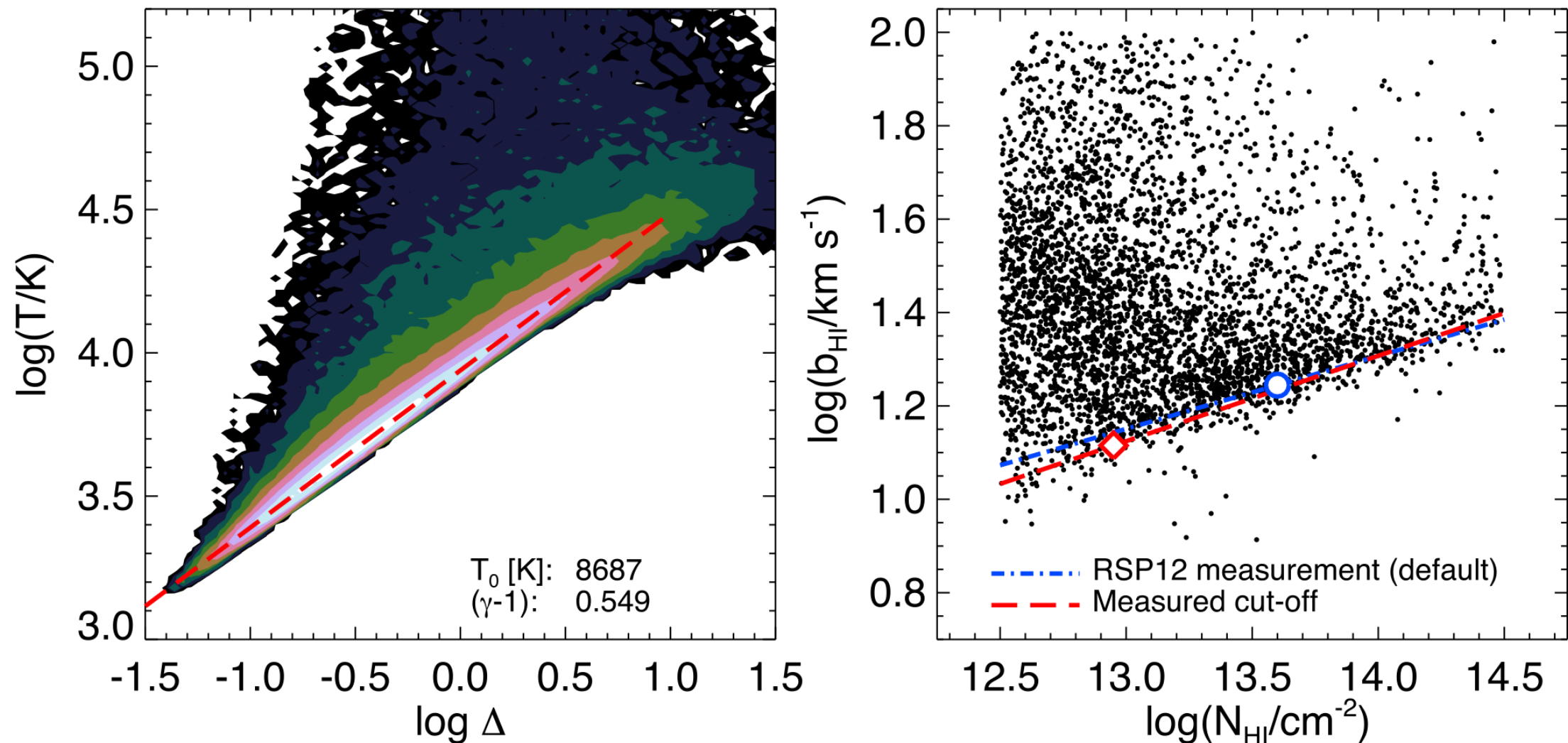
The temperature evolution at $\delta=0$



Becker et al. (2011)

- Evidence for heating; very likely extended He-II reionisation see also [Schaye et al. \(2000\)](#), [Lidz et al. \(2010\)](#), [Garzilli et al. \(2012\)](#)
- The slope of the temperature-density relation remains a systematic

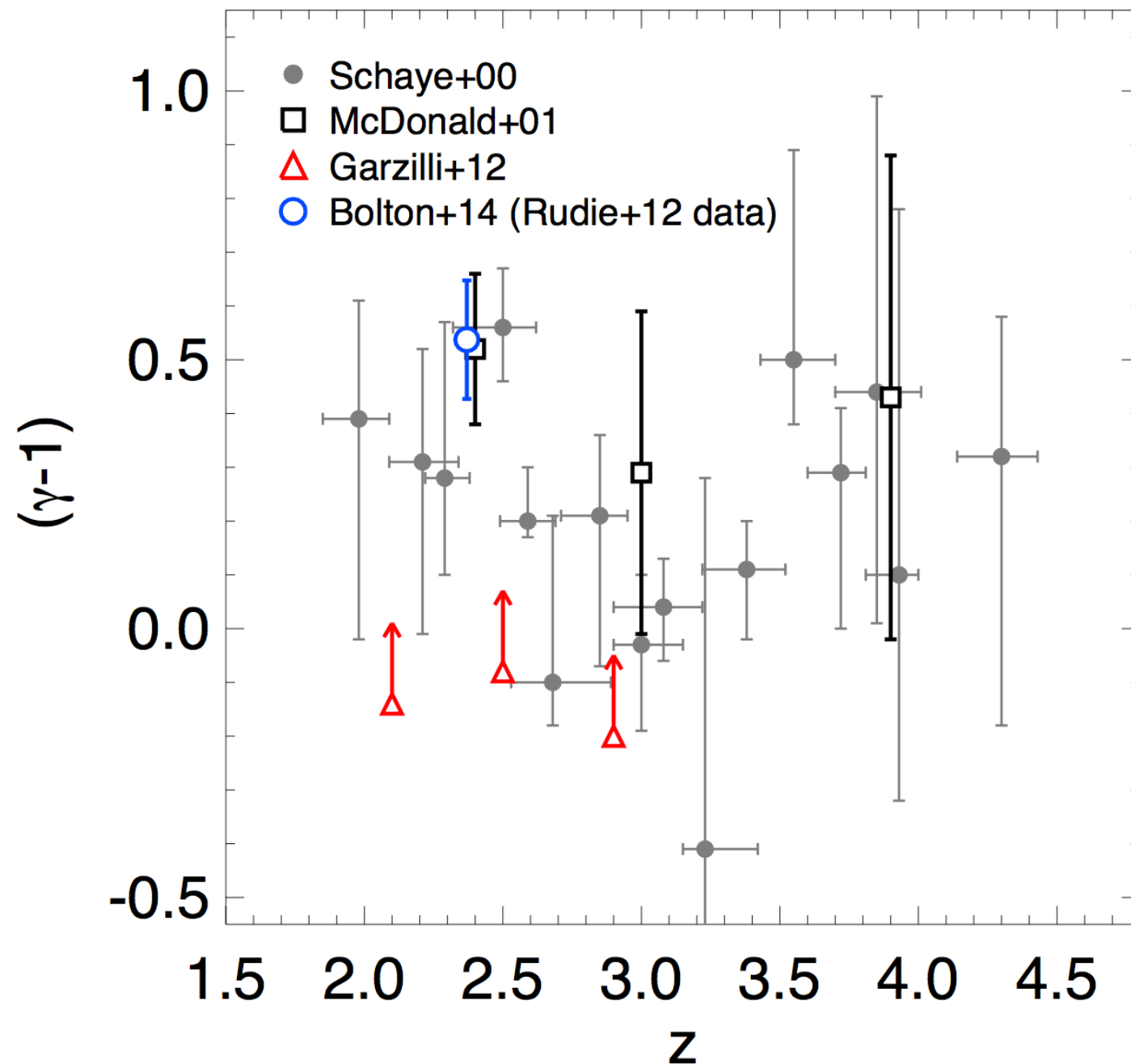
The b-N cut-off at $z=2.4$ revisited



Bolton et al (2014)

Re-analysis (with hydro simulations) of the b-N cut-off measured at $z=2.4$ from the KBSS (Rudie et al 2012, Schaye et al. 1999).

Temperature-density relation



Challenging to measure γ ;
narrow range of densities
probed by Ly- α forest

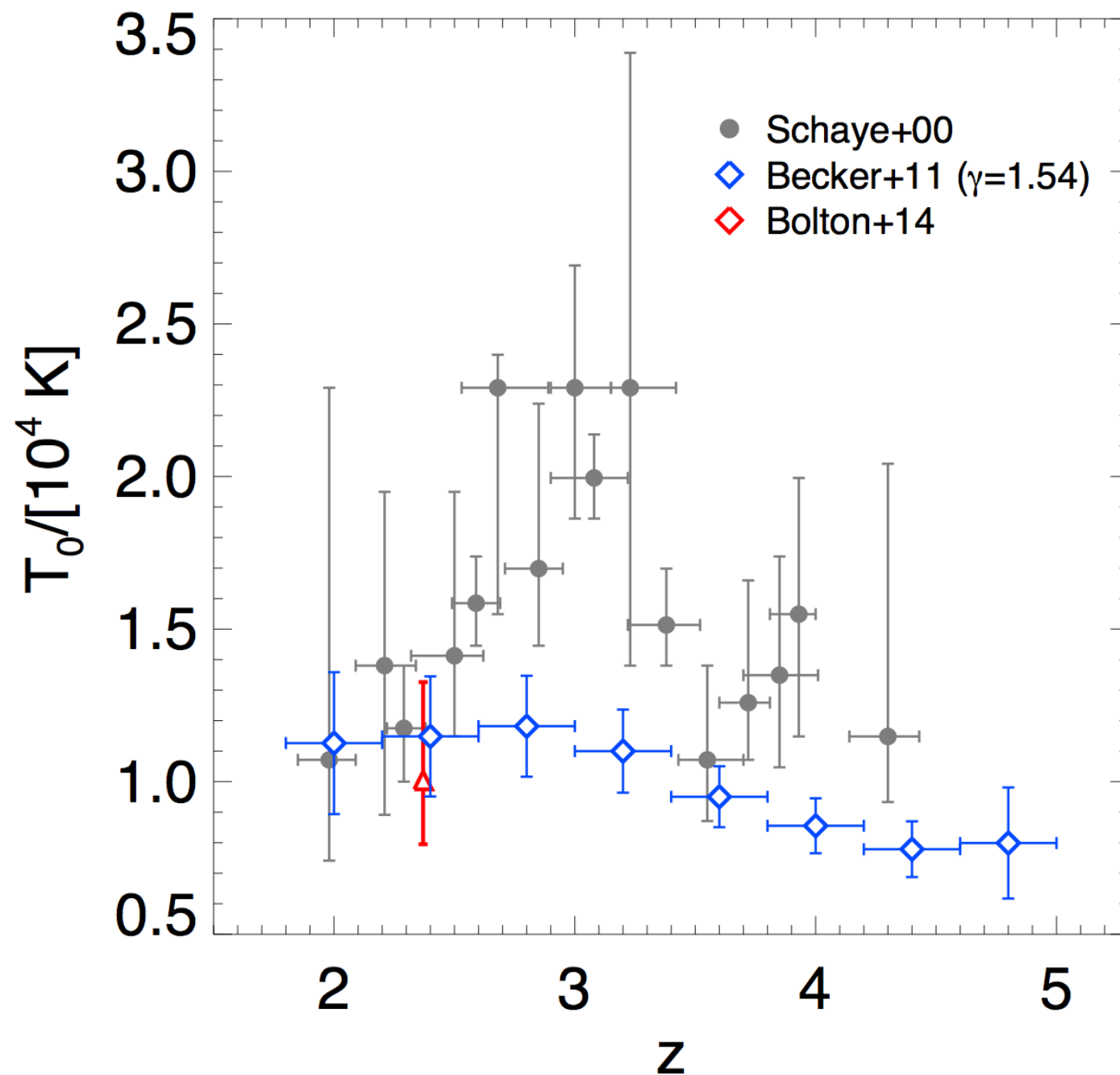
b-N cut-off from KBSS:

$$\gamma = 1.54 \pm 0.11$$

cf. [Rudie+12](#)

$$\gamma = 1.47 \pm 0.10$$

Consistency at $z=2.4$



Encouraging consistency
between curvature and b-N cut-
off at $z=2.4$

b-N cut-off from KBSS:

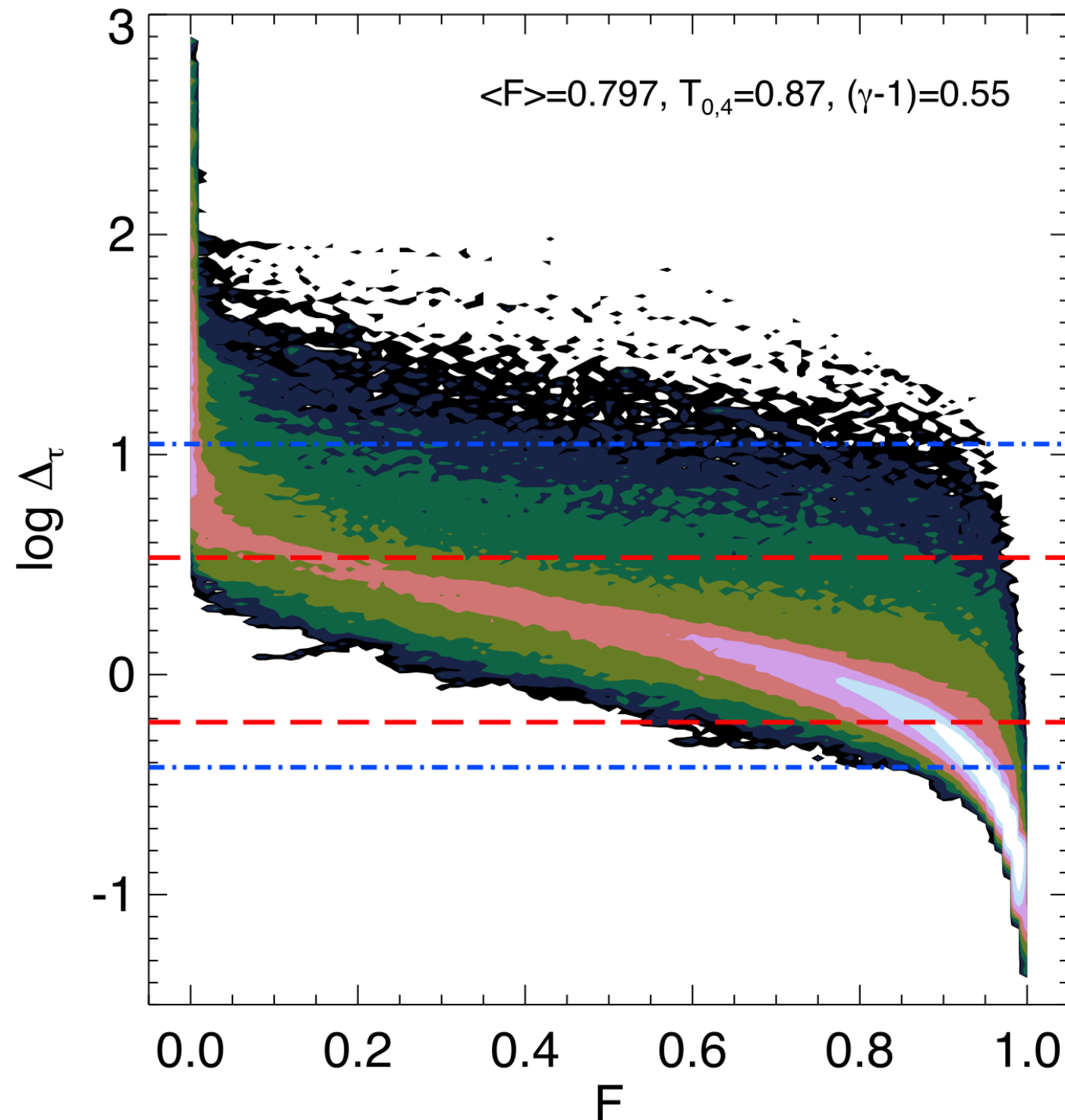
$$T_0 = 10000^{+3200}_{-2100} \text{ K}$$

cf. [Rudie+12](#)

$$T_0 = 18700 \pm 800 \text{ K}$$

Difference due to how N_{HI} at
 $\delta=0$ is determined in analytical
model

Thermal state of under-dense gas?

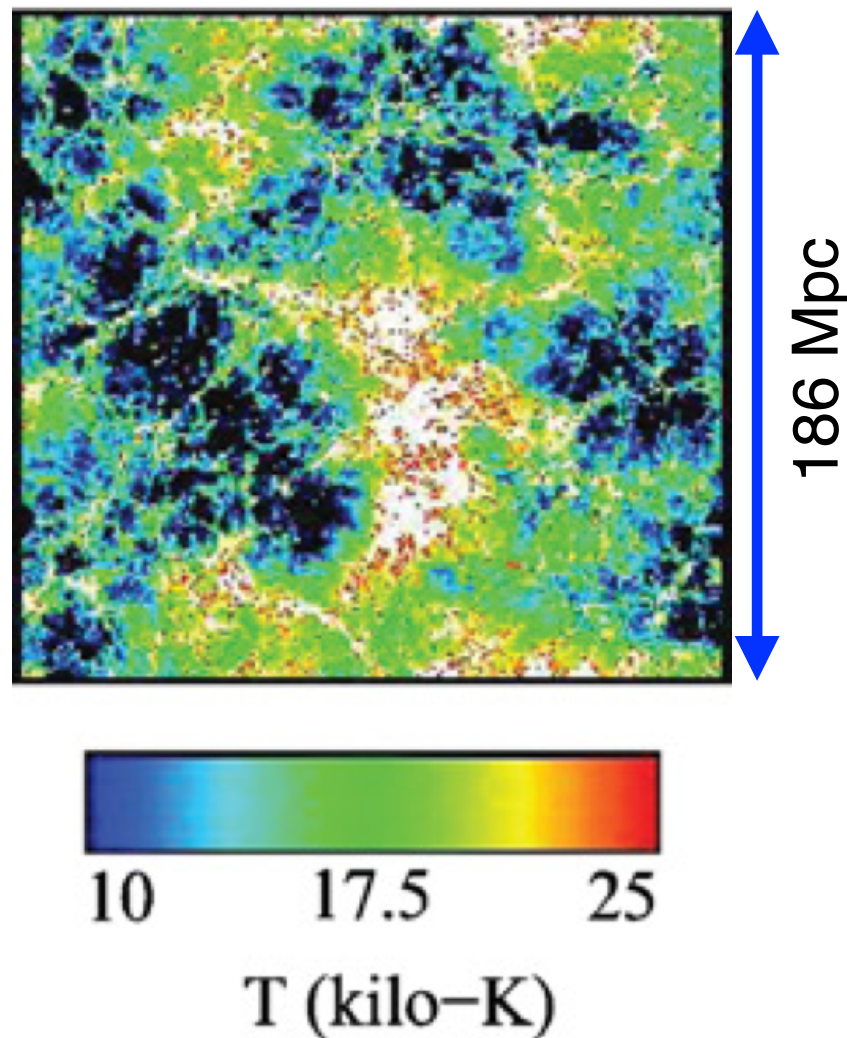


Bolton et al. (2014)

- Under-dense gas at $z=2.5$ is not well probed by the b-N cut-off, but dominates transmission at $F > 0.8$
- No reason to expect a tight power-law T - δ relation; inhomogeneous He-II reionisation will complicate matters.

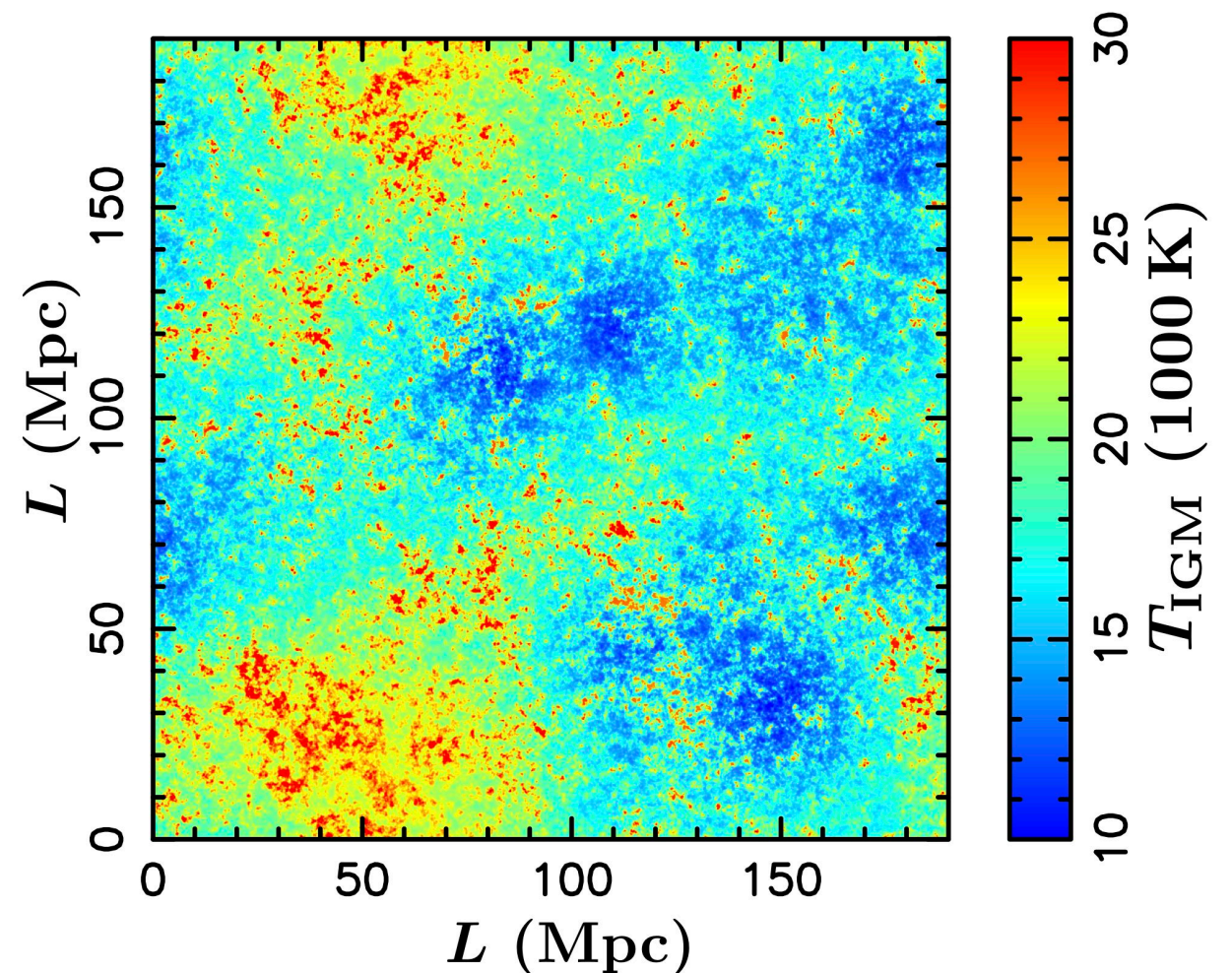
Temperature fluctuations at $z \sim 3$

Radiative transfer



McQuinn et al. (2009)

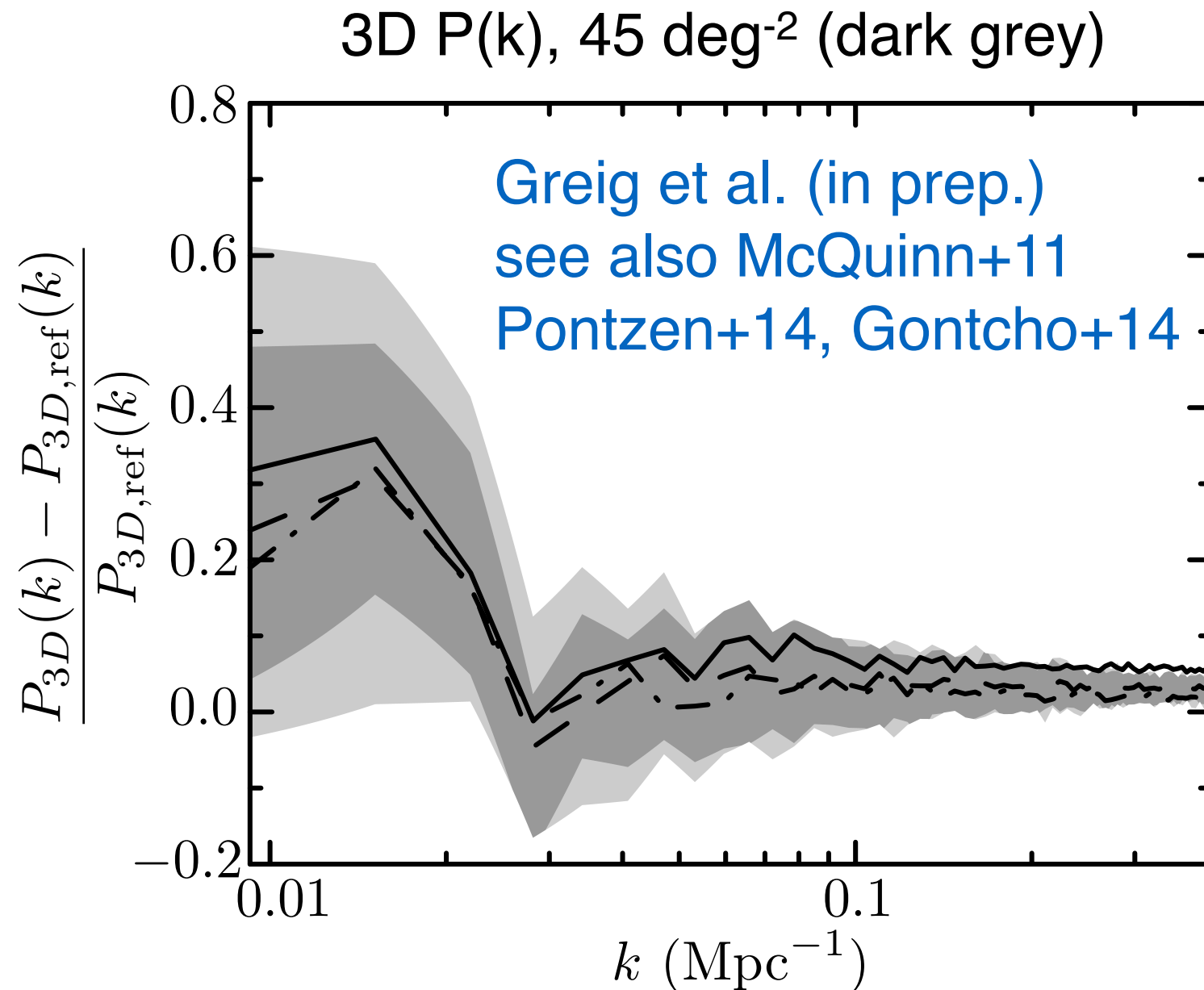
“Semi-numerical”



Greig et al. (in prep.); see also
Raskutti et al (2012), Lidz & Malloy (2014)

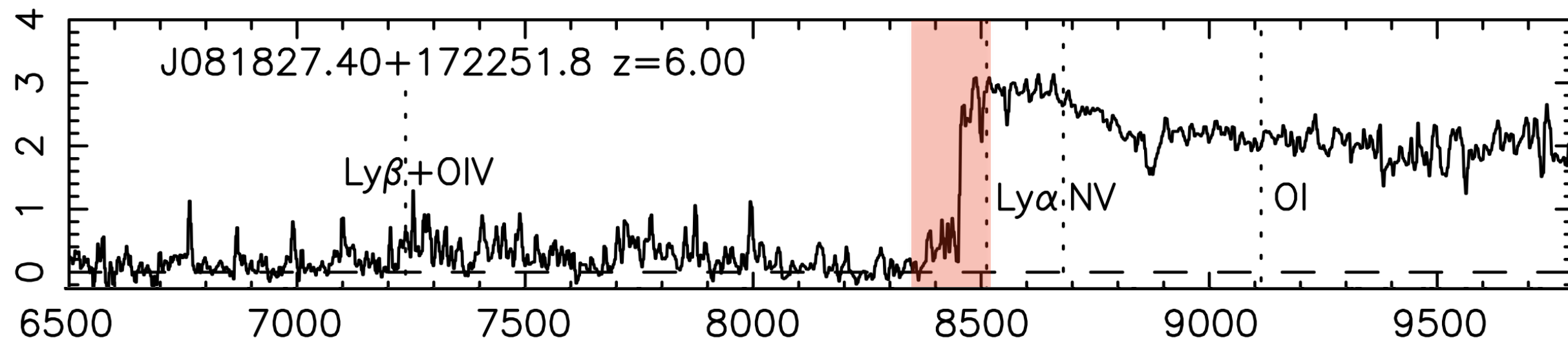
He-II reionisation will induce large scale (>30 cMpc) spatial fluctuations in the IGM temperature

Temperature fluctuations at $z \sim 3$



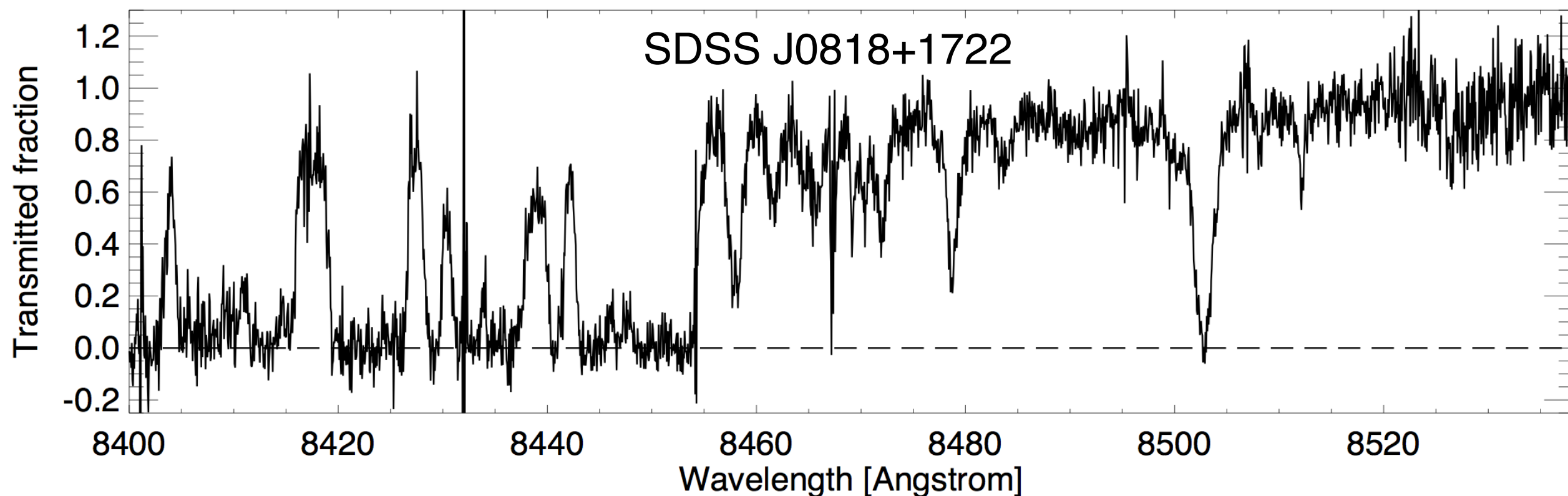
May be possible to detect >30 Mpc ionisation/temperature fluctuations
may be on large scales in the 3D power spectrum

The IGM temperature at $z \sim 6$



Keck/ESI, shown at $R \sim 1800$

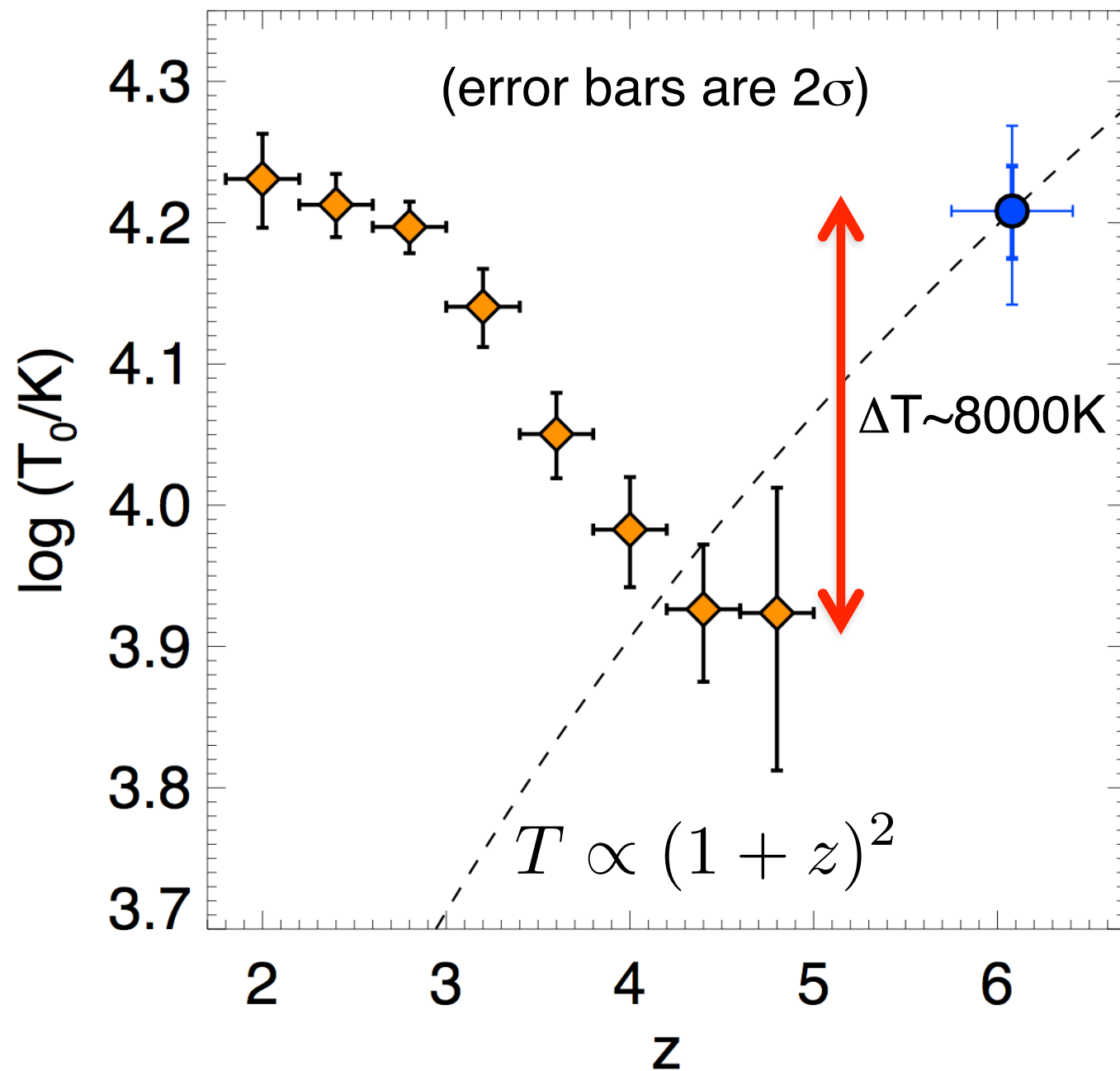
Fan et al. (2006)



Keck/HIRES, $R \sim 40,000$

Becker et al. (2007)

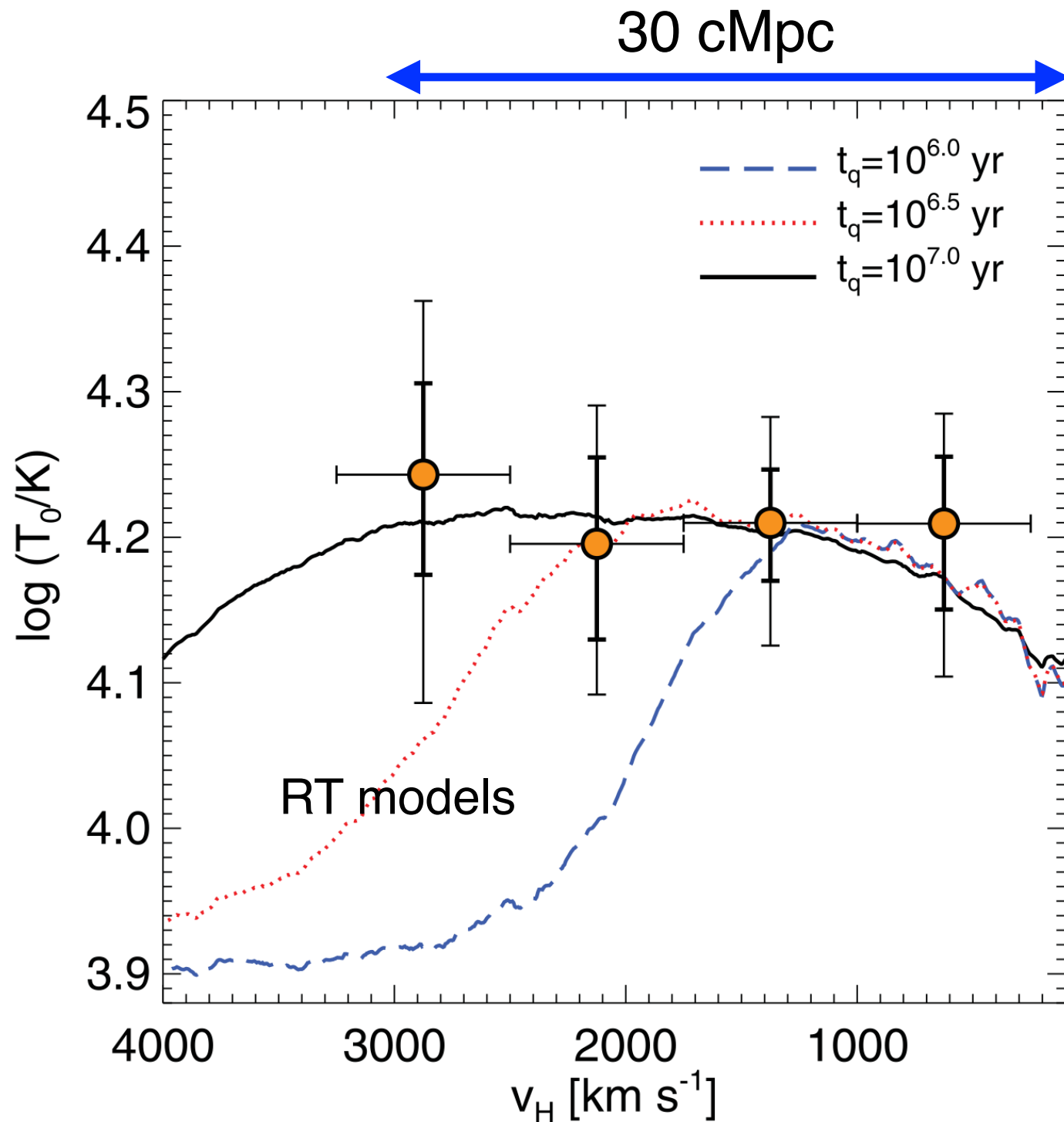
He-II photo-heating at $z \sim 6$



Bolton et al. (2012)

- 7 Keck/HIRES and Magellan/MIKE spectra
- Average temperature inferred at mean density $\sim 8000\text{K}$ hotter than general IGM at $z \sim 5$;
- Possible evidence for the initial stages of He-II reionisation within $\sim 30\text{ cMpc}$ of $z \sim 6$ quasars

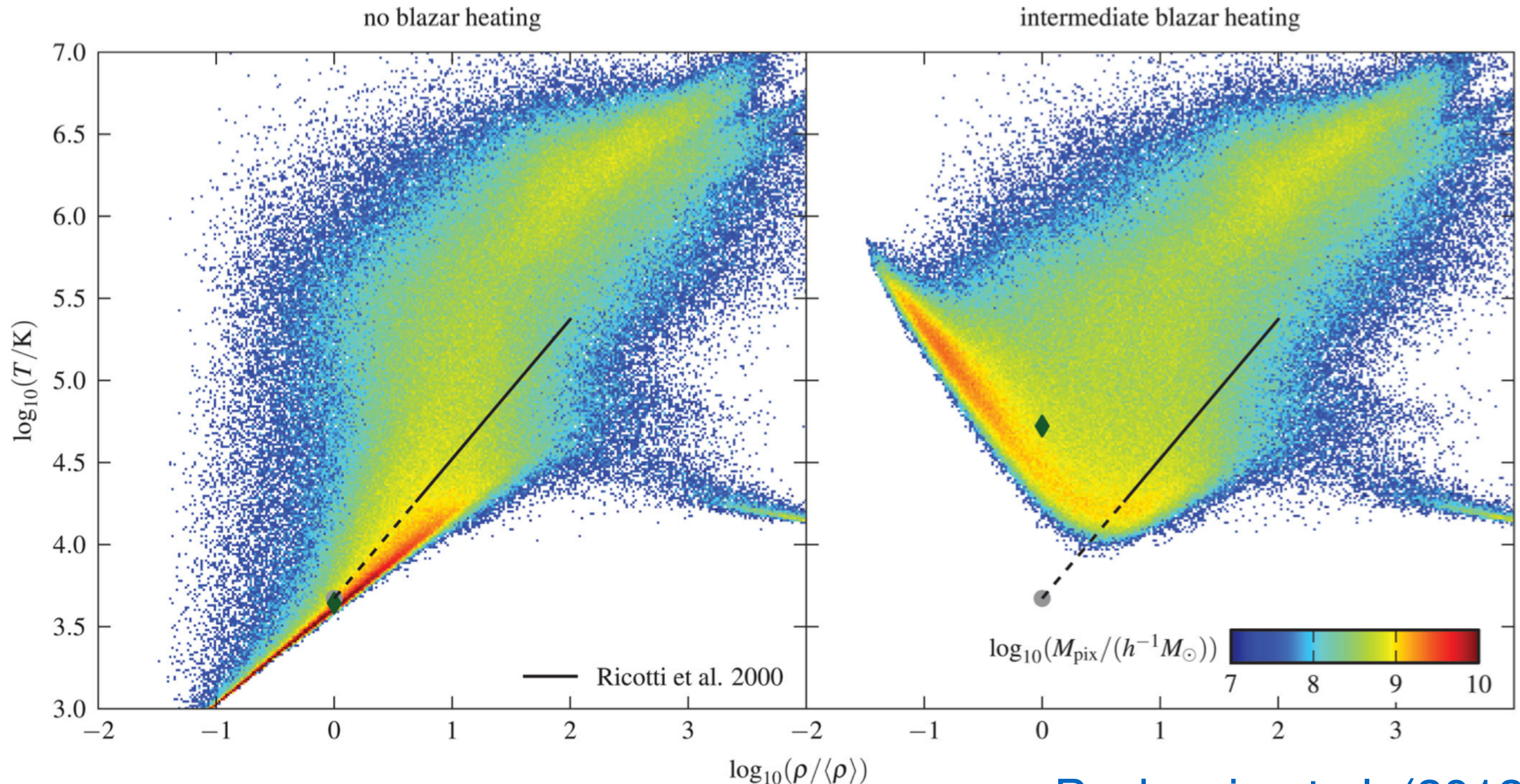
Thermal proximity effect?



Bolton et al. (2012)

- Average temperature as function of distance from quasars is constant out to 30 cMpc;
- Assuming quasars are photo-heating He-II in their vicinity, consistent with $t_Q > 3 \times 10^6$ yrs.

Other heating sources required?

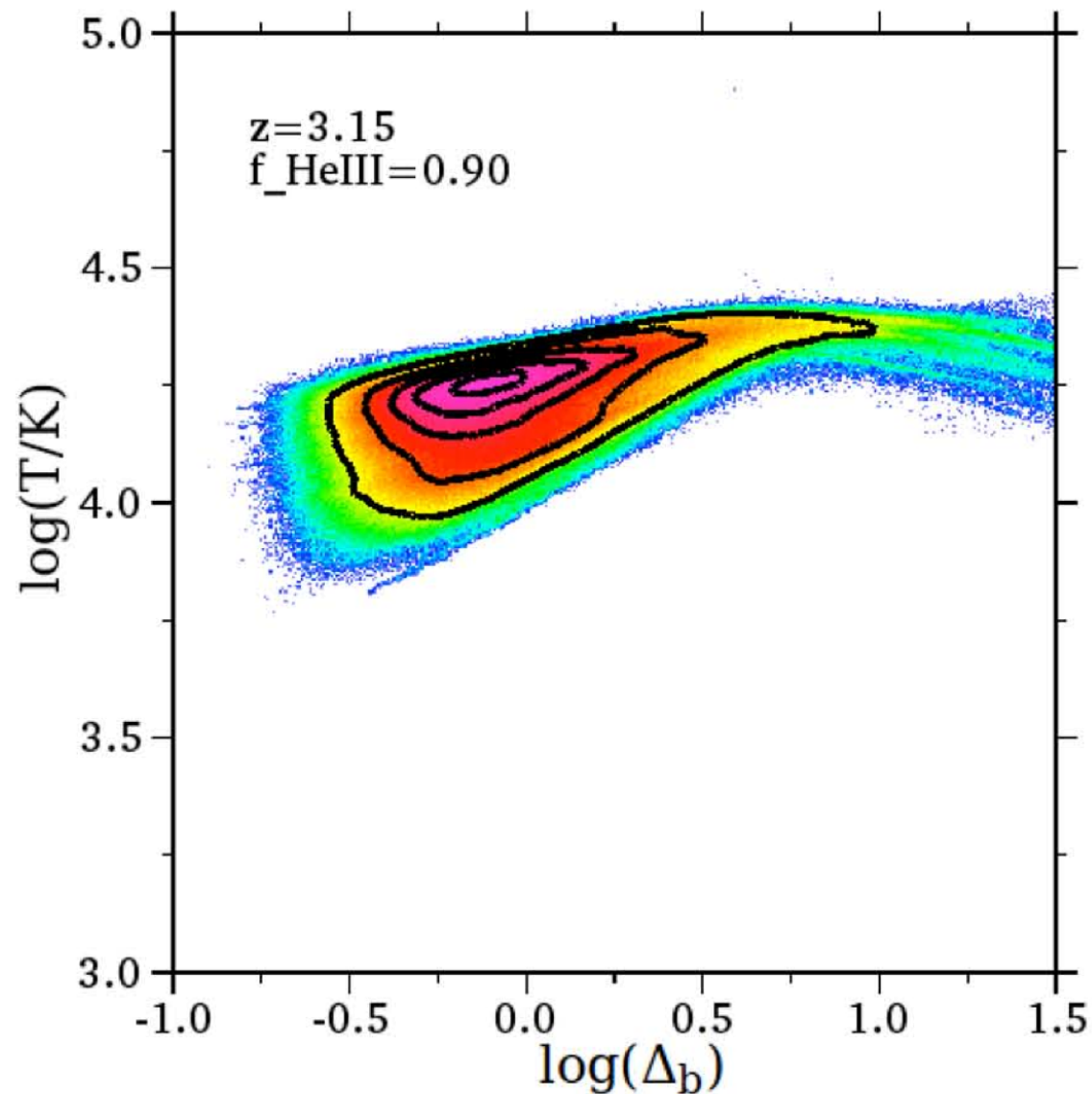


Puchwein et al. (2012)

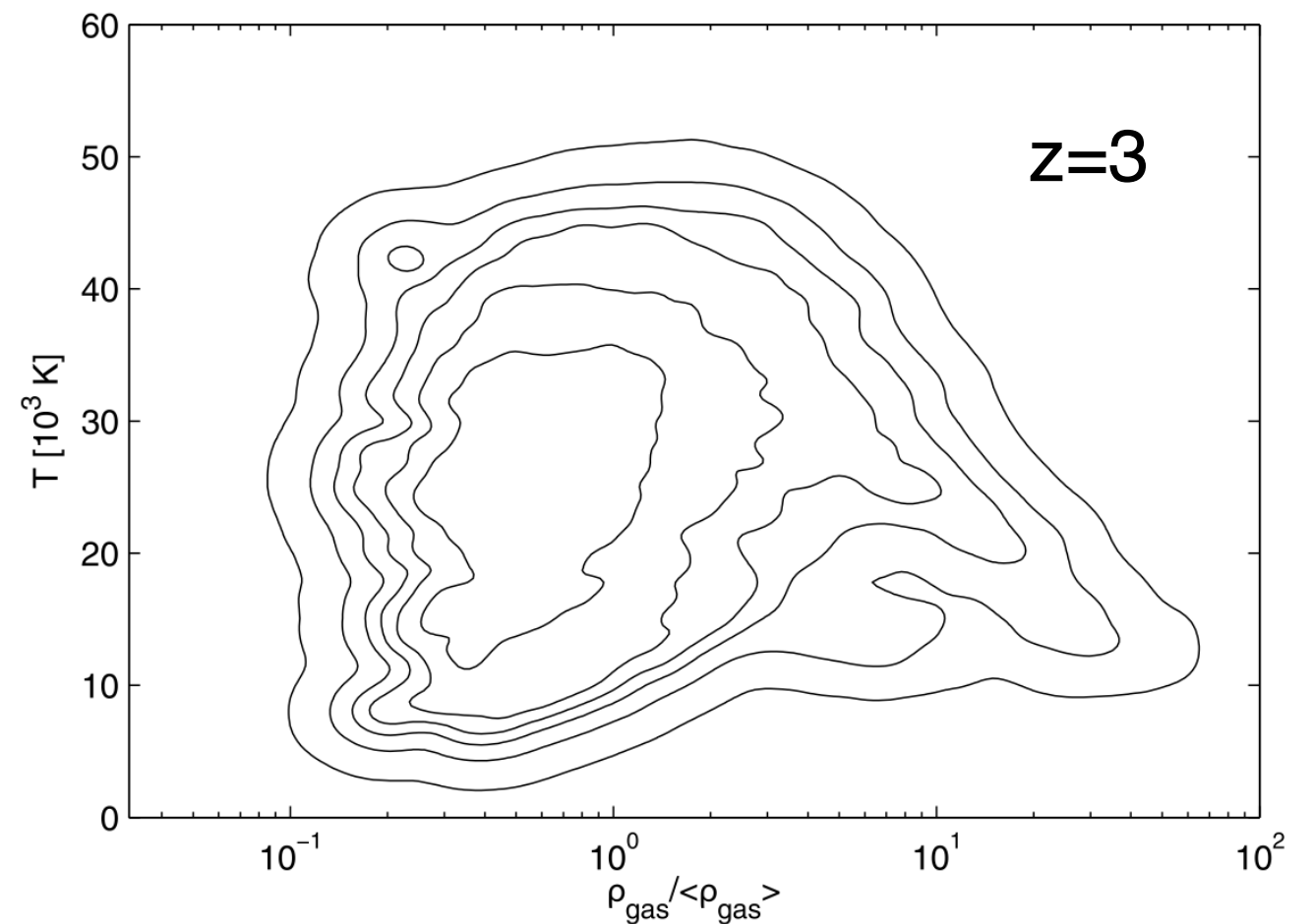
Volumetric heating from blazar TeV emission heats under-dense gas (but see [Miniati & Elyiv 2013](#), and [Kollmeier et al \(2014\)](#) for interpretation of $z\sim 0$ COS data in context of blazar heating)

Radiative transfer effects

Inhomogeneous heating and spectral filtering will induce scatter in the temperature-density relation during He-II reionisation



Compostella et al. (2013)



Meiksin & Tittley (2012)

Key results:

- $T(\Delta)$ now well measured from $2 < z < 5$ and around quasars at $z \sim 6$
- Consistency between two classes of methodologies at $z \sim 2.4$
- Temperatures consistent with extended He-II reionisation from $z \sim 4.5$, possibly starting as early as $z = 6$
- Spatial fluctuations in ionisation and temperature during He-II reionisation may be detectable with 3D $P(k)$ at $k \sim 0.01 - 0.03 \text{ Mpc}^{-1}$
- Inverted T - ρ relation unlikely at $z = 2.4$, but some caution required; the under dense IGM is not well probed by the forest at $z < 3$

Some open questions/challenges:

- Density dependence of IGM temperature, γ [b-N cut-off? alternatives?]
- Inferring the Jeans scale in the IGM [1d or 3d $P(k)$, QSO pairs?]
- Thermal history at $z < 2$, evidence for blazar heating [COS?]
- Spatial fluctuations in thermal state at $z \sim 3$ [3d $P(k)$, BOSS?]
- Thermal proximity effect at $z < 6$ [line widths?]
- Further comparison to radiative transfer + hydro required [simulations]

See the next talks!